

Asfar Hossain Sitab

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PROFESSIONAL SUMMARY

BSc (CSE) graduate of East West University (May 2026, CGPA: 3.71/4.00), currently pursuing an MSc in Artificial Intelligence and Machine Learning. Research expertise spans multi-task learning, computer-vision pipelines, and explainable-AI integrations on real-world climate, agricultural, and biomedical datasets. Skilled in deep-learning stacks (PyTorch, TensorFlow, YOLO) and full-stack development (PHP, Node.js, Docker), with a track record of high-impact academic projects and consistent Dean's and Trustees' Scholarship recognition. Eager to contribute innovative, data-driven solutions in both research and industry environments as an ML practitioner.

EDUCATION

MSc in Artificial Intelligence and Machine Learning <i>East West University</i>	<i>2026 – Present</i> Dhaka, Bangladesh
BSc in Computer Science and Engineering <i>East West University</i>	<i>2022 – May 2026</i> Dhaka, Bangladesh
Graduated CGPA: 3.71 / 4.00	
Higher Secondary Certificate (HSC) – Science <i>St. Gregory High School and College</i>	<i>2019 – 2020</i> Dhaka, Bangladesh
GPA: 5.00 / 5.00	
Secondary School Certificate (SSC) – Science <i>Trust School</i>	<i>2016 – 2018</i> Dhaka, Bangladesh
GPA: 5.00 / 5.00	

TECHNICAL SKILLS

Programming	Python, C, Java, PHP, JavaScript (ES6+), SQL, L ^A T _E X
ML / DL Frameworks	PyTorch, TensorFlow, Scikit-learn, CatBoost, LightGBM, OpenCV, YOLO (v8/v11/v12)
XAI & Analytics	SHAP, LIME, MODIS NDVI, NASA POWER, Remote Sensing, Statistical Analysis
Web & DevOps	Node.js, PHP 8, Docker, Apache, MongoDB, MySQL/SQLite, RESTful APIs
Platforms & Tools	GitHub, VS Code, Google Colab, Kaggle, CodeCarbon
Research Skills	Multi-task Learning, Ablation Studies, Technical Writing, Predictive Modelling

RESEARCH INTERESTS

Machine Learning & Deep Learning • Explainable AI (XAI) • Data Science & Predictive Modelling • Climate & Sustainability Analytics • Computer Vision • AI for Decision-Support Systems

ONGOING RESEARCH

ClimaDecideRisk: Multi-task Interpretable ML for South Asian Agriculture

Technologies: CatBoost, SHAP, NASA POWER, MODIS NDVI, Remote Sensing

- Developed a joint-modelling framework for climate volatility, decision dynamics, and psychological risk.
- Integrated open-source climate and remote-sensing data: 5,850 weekly observations across 8 districts.
- Achieved $R^2 = 0.988$ (FVI) and $R^2 = 0.975$ (PRPI) using CatBoost with SHAP-based interpretability.

TropiLite-IE: Lightweight AI for Smart Building Energy Prediction

Technologies: Python, GRU, LightGBM, SHAP, CodeCarbon

- Designing a hybrid GRU–LightGBM model for building energy consumption in tropical climates.
- Integrating physics-informed thermodynamic features (moist-air enthalpy) for predictive accuracy.
- Using CodeCarbon to quantify environmental impact; SHAP for model transparency.

Anomaly-Based Deep Learning for Zero-Day Attack Detection

Technologies: PyTorch, Network Security, Anomaly Detection

- Developing frameworks to detect unseen cyber-threats in network traffic using deep learning.
- Implementing anomaly-based learning on traffic datasets for intrusion detection and threat profiling.

RECENT PROJECTS (2024–2025)

DeepCarDamage: Automated Vehicle Damage Assessment Framework

Technologies: YOLOv8/11/12-Seg, MobileNetV2, CNN, CarDD Dataset

- Built a dual-pipeline architecture for automated detection, instance segmentation, and severity assessment.
- YOLOv12-Seg achieved 72.9% mAP@50 and 68.7% mask IoU for precise damage delineation.
- MobileNetV2 reached 70% severity-classification accuracy at 135 FPS for real-time appraisal.

Semi-Supervised Instance Segmentation of Surgical Anatomy

Technologies: YOLOv12-Seg, Mean Teacher, Pseudo-labelling, Dresden Dataset

- Evaluated SSL strategies (Mean Teacher, MixMatch, Pseudo-labelling) to reduce dense annotation needs.
- Mean Teacher proved most stable: higher coherent instance masks and fewer false positives.
- Demonstrated that YOLO-based SSL can scale real-time surgical-scene understanding effectively.

PREVIOUS PROJECTS

QuickBite – Online Food Ordering & Delivery Platform

Technologies: PHP 8.0, JavaScript (ES6+), MySQL/SQLite, Docker, Apache

- Developed an end-to-end platform connecting customers, restaurants, and delivery personnel.
- Engineered a complete management system for restaurant menus, order processing, and delivery logistics.

Fruit Quality Classifier – Data Science / Machine Learning

Technologies: Python, TensorFlow, Scikit-learn, SHAP, LIME

- Built multi-class fruit-quality classification using 10+ deep-learning models (VGG, ResNet, etc.).
- Implemented SHAP and LIME to interpret feature importance and visualise model decision-making.

EWU CLUB HUB – University Club Management Platform

Technologies: HTML5, CSS3, Node.js, MongoDB, RESTful APIs

- Created a centralised hub for university club activities with event management and notifications.
- Developed secure RESTful APIs managing club data with intuitive UX for students and admins.

ACHIEVEMENTS & SCHOLARSHIPS

- **Dean’s Scholarship** – 4th Year, East West University (2025–2026)
- **Trustees Scholarship** – East West University (2022–2026)
- **Merit Medals (Medha Lalon)** – East West University (2022–2025)

EXTRACURRICULAR ACTIVITIES

- **Debater:** Participated in national-level debates broadcast on BTV.
- **Chess Player:** 1st position in inter-college chess competition.

LANGUAGES

Bangla: Native proficiency • **English:** Professional proficiency

REFERENCES

Dr. Mohammad Manzurul Islam
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